

KUUSAMO, Finland

WMO: 028690

Lat: 65.996N

Lon: 29.223E

Elev: 866

StdP: 14.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-18.4	-30.3	1.0	-24.3	-23.6	1.5	-18.3	19.4	23.1	17.5	23.2	1.6	80	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	76.1	61.2	72.4	59.5	69.0	58.2	64.0	71.4	61.8	69.0	59.9	66.7	8.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	83.1	67.3	58.9	76.7	64.7	56.8	71.1	62.9	29.7	71.1	28.1	69.0	26.7	66.2	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.5	13.9	-30.4	80.4	6.5	3.7	-35.1	83.0	-38.9	85.1	-42.5	87.2	-47.2	89.8		
(5)			-30.5	66.5	6.3	2.5	-35.1	68.3	-38.8	69.8	-42.3	71.2	-46.9	73.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.9	10.3	11.2	18.5	30.1	41.4	52.5	58.8	54.1	45.0	33.6	22.6	15.4		
	DBStd	19.24	14.10	13.24	10.13	8.18	8.39	7.09	6.21	6.02	6.38	7.58	10.38	13.16		
	HDD50	6850	1231	1087	977	597	290	53	7	28	175	510	823	1072		
	HDD65	11738	1696	1507	1442	1047	730	380	209	342	601	974	1273	1537		
	CDD50	608	0	0	0	0	24	127	279	154	24	0	0	0		
	CDD65	23	0	0	0	0	0	4	16	3	0	0	0	0		
	CDH74	196	0	0	0	0	4	50	112	30	0	0	0	0		
	CDH80	18	0	0	0	0	0	5	10	3	0	0	0	0		
	WSAvg	6.3	5.8	6.2	6.6	6.3	6.8	6.7	6.1	5.8	6.4	6.5	6.3	6.4		
	PrecAvg	23.5	1.6	1.3	1.2	1.3	1.9	2.4	3.0	2.8	2.3	2.0	2.0	1.6		
	PrecMax	31.1	3.5	2.9	2.3	2.7	3.9	5.8	5.9	5.6	4.6	4.5	4.4	3.7		
(15) Precipitation	PrecMin	16.4	0.4	0.2	0.3	0.2	0.1	0.4	0.4	1.0	0.4	0.6	0.3	0.5		
	PrecStd	3.4	0.7	0.6	0.5	0.7	0.8	1.1	1.3	1.1	1.0	1.0	0.9	0.7		
	DB	35.3	37.1	42.2	56.1	73.5	79.7	81.2	77.8	65.7	51.7	43.5	37.3			
	MCWB	34.0	35.0	36.0	45.0	54.8	61.0	64.0	64.1	56.6	49.4	42.3	36.1			
	2% DB	32.3	33.7	37.5	49.4	67.2	73.7	77.1	72.4	60.7	48.3	39.8	34.6			
(21) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	31.8	32.6	33.4	41.0	53.2	58.7	62.1	60.6	54.0	46.5	38.9	33.7			
	5% DB	30.4	31.5	34.5	45.0	60.9	69.0	73.7	68.4	57.5	46.1	36.5	32.9			
	MCWB	29.9	30.7	31.8	38.1	49.7	56.3	60.8	59.0	52.3	44.5	35.8	32.3			
	10% DB	28.1	28.7	32.2	41.9	55.9	65.3	70.2	64.8	55.1	43.6	34.1	31.4			
	MCWB	27.6	27.9	30.4	36.3	46.7	54.3	59.6	57.3	51.5	42.3	33.5	30.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	34.0	35.0	36.9	45.6	57.6	64.5	67.4	65.4	58.1	49.7	42.7	36.1			
	MCDB	34.8	36.3	41.0	53.3	67.4	72.9	76.1	74.4	63.2	50.8	43.2	36.6			
	2% WB	32.0	32.6	34.1	42.2	54.3	61.2	65.1	62.7	55.4	47.0	39.0	33.8			
	MCDB	32.3	33.4	36.4	47.7	66.0	70.2	72.6	69.8	59.1	48.1	39.7	34.3			
	5% WB	30.0	30.6	32.3	39.1	50.9	58.5	62.9	60.2	53.5	44.5	35.8	32.3			
(31) Mean Daily Temperature Range	MCDB	30.4	31.3	34.0	43.8	60.2	67.5	70.2	65.9	56.8	45.6	36.3	32.7			
	10% WB	27.4	27.9	30.5	36.7	47.5	55.8	60.9	58.3	51.7	42.3	33.6	30.6			
	MCDB	27.9	28.5	32.1	40.6	54.3	62.6	67.9	63.5	54.5	43.4	34.0	31.1			
	5% DB	13.2	13.8	17.2	15.8	15.4	15.3	14.9	14.4	11.8	7.8	9.4	11.5			
	MCDBR	10.7	10.0	15.8	19.9	24.6	21.8	20.6	20.2	16.1	9.3	7.4	8.4			
(36) Clear-Sky Solar Irradiance	MCWBR	10.7	9.5	13.0	12.7	13.1	9.9	8.6	10.1	10.2	8.0	7.4	8.3			
	5% WB	10.7	9.7	12.8	17.4	22.8	19.1	16.6	16.3	13.8	8.3	7.2	8.3			
	MCDBR	10.8	9.4	10.9	11.5	12.9	9.9	8.6	9.5	9.6	7.8	7.2	8.2			
	taub	0.194	0.262	0.295	0.327	0.339	0.339	0.357	0.336	0.312	0.284	0.351	0.194			
	taud	1.926	2.107	2.160	2.247	2.453	2.489	2.458	2.491	2.540	2.363	2.700	1.950			
(42) All-Sky Solar Radiation	Ebn at Noon	78	188	241	264	274	277	267	262	244	187	81	11			
	Edh at Noon	6	17	27	33	30	30	30	26	20	14	6	1			
	RadAvg	29	187	648	1210	1500	1606	1500	1093	587	196	44	5			
(45)	RadStd	2	36	60	118	141	119	151	109	61	31	7	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page